Request formation

Base rs1
I / S immediate

ADD

Width / alignment
Byte / half / word

Store rs2

Store alignment
Byte enables[3:0]

Load /
store /
cache op

Request
registers
Address / data
Strobes /
attributes

Memory interface

D-Cache / TCM

Request / accept
Read data / ack / error

Request and completion state

Request & accept

Pending bit
Accept: set Ack: clear

pending AND NOT ack_ok
Delay / stall / hold

Unaligned
E1 → E2
Synthetic response

Push: accept OR (unaligned E1 AND NOT delay)

Completion: ack / error / unaligned

Response formatting

Metadata FIFO
2 × 36 bit
Address32 + flags4

Byte / half / word
Selection

Sign / zero
Extension

Value

Writeback
Valid / fault

Fault address